Hang Zhang

hangzhang@ethz.ch

EDUCATION

Wuhan University, China — 09/2012-07/2017
Bachelor of Architecture

University of Pennsylvania, USA — 09/2018-12/2019
Master of Science in Advanced Architectural Design

ETH Zurich, Switzerland — 09/2021-10/2026
PhD

PRACTICES

CISDI Engineering Co., Ltd — 08/2017-05/2018
Intern Architect: Primary Analysis and Design

University of Pennsylvania — 01/2020-08/2020
Research Assistant / Teaching Assistant

Roboticplus.AI — 09/2020-05/2021
Algorithm Developer: Robotic Timber Fabrication

FEATURED PUBLICATIONS

MaskPLAN: Masked Generative Layout Planning from Partial Input — 2024
Hang Zhang, Anton Savov, Benjamin Dillenburger
Published in **CVPR 2024**

Text-conditioned irregular floorplan generation via composable inpainting diffusion — 2026
Hang Zhang, Anton Savov, Benjamin Dillenburger
Published in **Automation in Construction**

Masked irregular layout planning from Partial Input with Fused Modalities — 2026
Hang Zhang, Anton Savov, Benjamin Dillenburger
Published in **Automation in Construction**

FreeLayout: Free Range Multi-modal input to Free Form Floorplan Layout Generation — 2026
Hang Zhang, Anton Savov, Benjamin Dillenburger
Published in **Advanced Engineering Informatics**

IN PROGRESS

Fabrication-Aware Floorplan Generation via Bootstrapped Label-Free Supervision and Constructed Diffusion Conditioning — 2026
Under Review: **Automation in Construction**
Hang Zhang, Jianpeng Cao, Anton Savov, Daniel Hall, Benjamin Dillenburger

Single-material 4D-printed shape-morphing structures via spatially patterned strain trapping — 2026
Under Review: **Advanced Functional Materials**
S M Asif, Hang Zhang, Lin Yang, Aoyi Luo, Joseph D. Paulsen, James H. Henderson

FabVLM: Efficient Training of Multi-Agent Vision-Language Model to Fabrication-Aware Generative Design 2026
To be Submitted
Hang Zhang, Anton Savov, Daniel Hall, Benjamin Dillenburger

OTHER PUBLICATIONS

3D Model Generation on Architectural Plan and Section Training through Machine Learning 2019
Hang Zhang
Published on **MDPI Technology**

3D Architectural Form Style Transfer through Machine Learning 2020
Hang Zhang, Ezio Blasetti
Published on **CAADRIA 2020**

Machine Learning Aided 2D-3D Architectural Form Finding at High Resolution 2020
Hang Zhang, Ye Huang
Published on **CDRF DigitalFuture 2020**

Text-to-Form 3d Prediction by Linguistic Descriptions 2020
Hang Zhang
Published on **ACADIA 2020**

Energy-Aware Design Predicting Building Performance from Layout Graphs 2022
Jianpeng Cao, Hang Zhang, Anton Savov, Daniel Hall, Benjamin Dillenburger
Published on **EC3 2022**

Machine Learning-based Optimization of the Design of Composite Pillars for Dry Adhesives 2022
Aoyi Luo, Hang Zhang, Kevin T.Turner
Published on **Extreme Mechanics Letters**

A Framework for the Design and Evaluation of Architectural Tilesets 2023
Hanbing Zhao, Anton Savov, Hang Zhang, Benjamin Dillenburger
Published on **ECAADE 2023**

Identifying the Design Feature That Causes Project Delay in DfMA: A Dominant Element Analysis Method for Project Scheduling 2023
Jianpeng Cao, Hang Zhang, Bo Pan, Ranjith Soman, Anton Savov, Daniel Hall
Published on **Computing in Civil Engineering 2023**

Fabrication-Aware Generative Design for Hybrid Prefabrication Using Stable Diffusion XL 2026
Hang Zhang, Jianpeng Cao, Anton Savov, Benjamin Dillenburger
Published on **EC3 2026**

SERVICES

Architecture Conference Reviewer: ACADIA, CAADRIA, CDRF

Architecture Journal Reviewer: Building and Environment
. Automation in Construction

Computer Science Conference Reviewer: CVPR, ICCV, ECCV
. Siggraph, Siggraph Asia
. NeurIPS, ICML

TEACHING

UPenn Master Course 2020
Form and Algorithm
Teaching Assistant

ETH Bachelor Course 2021–Present
Computational Design
Teaching Tutor

ETH Master Course 2022–2023
Advanced Computational Design
Teaching Tutor

ETH Master Thesis 2021–2024
MAS DFAB T3 Thesis
Thesis Supervisor

Workshop 2023
DigitalFUTURE RL-FabDesign Workshop
Host

Workshop 2024
ETH Seminar Week 7DayHouse Workshop
Co-host